

Integrated ROV-Deployable Microplastic Capture Cube

Collaborative DEP–NMP–ASW Filtration System

Technical & Development Brief

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1. Purpose & Context

This document outlines the **technical architecture, operational logic, and phased development pathway** of a modular, ROV-deployable microplastic capture cube.

The system is being presented privately to an interdisciplinary academic panel as part of an **Innovation & Application (IA) workshop**, and is intended to demonstrate:

- Technical coherence
- Scientific plausibility
- Development realism
- Readiness for collaborative research

2. Environmental Problem

Microplastic contamination persists because:

- Particle sizes range from microns to millimetres
- Densities vary widely (PE, PP, PET, fibres, composites)
- Many particles remain neutrally buoyant
- Passive filtration clogs or underperforms in moving water

No single physical mechanism reliably captures the full spectrum of microplastics in real-world aquatic environments.

3. Core Design Principle

The system is designed around **sequential particle manipulation**, not brute-force filtering.

Condition → Aggregate → Retain

This principle underpins the integration of three complementary technologies.

4. Technology Stack

4.1 Dielectrophoresis (DEP)

Role: Particle Conditioning

DEP applies non-uniform electric fields to induce motion in particles based on dielectric contrast.

In this system, DEP is used to:

- Reduce chaotic suspension
- Alter particle trajectories
- Bias microplastics toward aggregation zones

DEP operates continuously at low power and does not rely on direct contact or membranes.

4.2 Acoustic Sound Waves (ASW)

Role: Particle Aggregation

Controlled acoustic fields generate radiation forces that encourage suspended particles to cluster.

In this system, ASW is used to:

- Increase local microplastic concentration

- Promote aggregation of conditioned particles
- Improve downstream capture efficiency

Acoustic parameters are tunable based on water chemistry and flow conditions.

4.3 Nano-Magnetic Particles (NMP)

Role: Final Capture & Retention

Functionalised nano-magnetic particles bind to or physically entrap aggregated microplastics.

In this system, NMPs enable:

- Magnetic retention without fine mesh filtration
- Cartridge-based capture modules
- Safe recovery and processing of collected material

5. Why the Technologies Work *Together*

- DEP alone moves particles but does not retain them
- ASW alone clusters particles but does not extract them
- NMP alone requires sufficient particle concentration

When combined sequentially:

Stage	Mechanism	Outcome
1	DEP	Behavioural conditioning
2	ASW	Concentration & aggregation
3	NMP	Physical capture & retention

The system therefore **amplifies capture probability**, rather than relying on a single mechanism to do everything.

This collaborative architecture is the core innovation.

6. Physical Architecture

- Modular cubic housing (ROV-compatible)
- Internal flow paths designed for staged processing
- Replaceable capture cartridges
- Low-power onboard electronics
- Pressure-tolerant enclosure
- Designed for additive manufacturing and rapid iteration

The cube is **deployable, detachable, and retrievable** via ROV without diver intervention.

7. Phase-Based Development Pathway

Phase 1 — Mechanical & Systems Integration (Current)

- Cube geometry finalised
- Internal staging zones defined
- Cartridge concept validated
- Electronics architecture outlined
- Fabrication via 3D printing

Objective: Demonstrate physical feasibility and modularity.

PHASE 2A — CONTROLLED LABORATORY VALIDATION

Integrated DEP–ASW–NMP Microplastic Capture System

Project Lead: Rays Browns Environmental International Ltd

Application Context: Marine, riverine, and mining-impacted waters

Review Audience: Academic, Institutional, Governmental, Investment

Confidentiality: Pre-publication / Private Review

1. Purpose of Phase 2A (Scope Lock)

Can DEP, Acoustic Sound Waves, and Nano-Magnetic Particles operate sequentially within a shared water column to measurably increase microplastic capture efficiency compared to any single method alone?

Phase 2A **does not:**

- optimise capture rates,
- claim field readiness,
- or validate long-term environmental deployment.

It **does:**

- validate physical compatibility,
- establish interaction order,
- quantify incremental gains,

- and justify prototype escalation.

2. System Logic: Why These Technologies Can Work Together

The system follows a **three-stage functional pipeline**, not parallel interference.

Stage 1 — Dielectrophoresis (DEP): *Conditioning*

- Applies non-uniform electric fields
- Exploits dielectric contrast between microplastics and water
- Alters particle trajectories and orientation
- **Goal:** reduce stochastic motion and introduce deterministic behaviour

DEP is not used for final capture — only **pre-conditioning**.

Stage 2 — Acoustic Sound Waves (ASW): *Aggregation*

- Applies standing or travelling acoustic fields
- Induces acoustic radiation forces
- Causes conditioned particles to cluster
- **Goal:** increase effective particle size and density

ASW does not bind particles — it **makes binding possible**.

Stage 3 — Nano-Magnetic Particles (NMP): *Retention*

- Functionalised magnetic particles introduced downstream
- Preferentially adhere to aggregated plastic clusters
- Captured via removable magnetic cartridge
- **Goal:** physical extraction without dispersal

NMPs are passive until aggregation has occurred.

Key Insight (Critical for Review Panels)

Each technology **fails alone** for known reasons.

Each technology **enables the next** when sequenced correctly.

3. Phase 2A Experimental Geometry

Water Column:

- Transparent vertical or horizontal chamber
- Controlled flow (laminar to low-turbulence)
- Modular insertion points for DEP electrodes and acoustic transducers

Control Variables:

- Salinity
- Temperature
- Particle size distribution
- Flow velocity
- Field strength (electric + acoustic)

Materials:

- Synthetic microplastics (PE, PP, PET fragments)
- Magnetite-based or coated NMPs
- Inert carrier fluid for calibration runs

4. Experimental Design

4.1 Control Runs

- No DEP / No ASW / No NMP
- Single-technology runs
- Dual-technology permutations

4.2 Integrated Runs

- DEP → ASW → NMP (locked order)
- Repeated trials ($n \geq 3$ per condition)

4.3 Measured Outputs

- Particle concentration (pre/post)
- Aggregation size distribution
- Magnetic recovery mass
- Energy input per capture unit

5. Success Criteria (Phase 2A)

Phase 2A is considered successful if **all** are true:

1. No destructive interference between fields
2. Aggregation measurable after DEP + ASW
3. Magnetic retention exceeds ASW-only baseline
4. Capture repeatable across trials
5. No uncontrolled nanoparticle dispersion

6. Roles & Responsibilities (Pre-Funding Safe)

- **Electronics:** built in-house (DEP drivers, acoustic control)
- **Mechanical:** 3D-printed housings and flow chambers
- **Laboratory Access:** partner institutions (post-funding)
- **Data Analysis:** academic collaborators
- **Environmental Oversight:** NGO consultation (non-lab stage)

This phase **does not require permanent lab residency**.

7. Phase 2A → Phase 2B Transition Gate

If Phase 2A confirms:

- compatibility,
- sequential amplification,
- and controllable retention,

Then Phase 2B proceeds to:

- integrated sealed prototype,
- higher flow rates,
- cartridge redesign,
- and pre-field stress testing.

PHASE 3 – FIELD PILOT DEPLOYMENT & VALIDATION

Project: Integrated ROV-Deployable Microplastic Capture Cube

Technology Stack: DEP · NMP · ASW

Organisation: Rays Browns Environmental International Ltd

Status: Pre-Commercial Field Validation

1. Phase 3 Purpose

Phase 3 transitions the system from controlled prototype testing into **real-world environmental operation**.

The objective is to **validate performance, durability, and data integrity under natural hydrodynamic, chemical, and biological conditions**.

This phase is explicitly **non-commercial** and designed for:

- Academic scrutiny
- Regulatory confidence building
- Investment risk reduction

2. Core Questions Phase 3 Must Answer

1. Can DEP, ASW, and NMP operate **collaboratively** in uncontrolled water bodies?
2. Does the system capture **real environmental microplastics**, not lab proxies?
3. Can the system maintain **operational stability** over multi-week deployments?
4. Is the data produced **defensible, reproducible, and auditable**?

3. Deployment Architecture

Deployment Method

- ROV-assisted placement of capture cube
- Autonomous submerged operation
- Timed retrieval via ROV or tethered lift

Operational Duration

- Initial deployments: **14–28 days**
- Iterative redeployments following inspection

Depth & Environment

- Freshwater river systems
- Estuarine / coastal waters
- Sediment-rich mining-impacted waters

4. Site Selection Strategy (Initial Pilots)

Site	Purpose
Waters of Leith (Scotland)	Urban freshwater validation, academic oversight
Ghana (illegal mining-impacted waters)	High sediment & heavy metal context
Estuarine / coastal site	Salinity gradient & mixed polymer exposure

Each site is selected to stress a **different failure mode**.

5. Technology Operation in Field Conditions

Dielectrophoresis (DEP)

- Conditions and polarizes microplastics using non-uniform electric fields
- Acts as the *primary particle discriminator*

Acoustic Sound Waves (ASW)

- Aggregates conditioned particles

- Increases effective capture size and residence time

Nano-Magnetic Particles (NMP)

- Bind aggregated plastics
- Retained in removable cartridge modules

Critical Principle:

Condition → Aggregate → Retain

6. Instrumentation & Data Collection

Metrics Collected

- Capture mass (dry & wet weight)
- Polymer identification (post-recovery lab analysis)
- Particle size distribution
- Energy consumption per m³ processed
- Cartridge saturation rates
- Fouling and degradation indicators

Data Handling

- Timestamped logs
- Site-specific metadata
- Chain-of-custody for recovered samples

7. Risk & Failure Mode Documentation

Phase 3 explicitly documents:

- Partial system degradation
- Reduced efficiency zones
- Biofouling impacts
- Sediment overload thresholds
- Power drift or acoustic attenuation

8. Phase 3 Deliverables

1. Field-verified performance dataset
2. Retrieved microplastic samples for independent lab analysis
3. ROV visual documentation (deployment, operation, recovery)
4. Engineering refinement recommendations
5. Regulatory-ready technical evidence package

9. Exit Criteria (Phase 3 → Scale Readiness)

Phase 3 is considered successful if:

- Capture is demonstrated across ≥ 2 environments
- No catastrophic system failures occur
- Data is accepted by at least one academic partner for review

Phase 4 — Scale-Up & Multi-Unit Array Deployment

Status: Planned

Dependency: Successful completion of Phase 3

Purpose: Transition from pilot units to scalable, networked remediation infrastructure.

4.1 Phase 4 Objectives

Phase 4 focuses on **capacity, repeatability, and economics**.

Primary objectives:

- Validate performance across **multiple cubes operating simultaneously**
- Establish array-level capture efficiency and redundancy
- Demonstrate manufacturability using low-complexity fabrication

- Quantify cost-per-kg microplastic removed
- Prepare system for government and institutional adoption

4.2 System Architecture — Multi-Unit Arrays

4.2.1 Array Configuration

Cubes are deployed in **modular clusters**, not fixed grids.

Typical configurations:

- Linear river arrays
- Radial estuary clusters
- Distributed seabed nodes
- Floating-subsurface hybrid systems

Each cube remains **individually autonomous**, preventing single-point failure.

4.3 Collaborative Technology Operation at Scale

System Interaction at Array Level

1. **ASW (Acoustic Sound Waves)**
 - a. Localised to each cube
 - b. No inter-cube interference due to frequency separation
 - c. Promotes aggregation before capture
2. **DEP (Dielectrophoresis)**
 - a. Operates internally per unit
 - b. Electric field isolation prevents cross-talk
 - c. Maintains particle conditioning consistency
3. **NMP (Nano-Magnetic Particles)**
 - a. Fully contained cartridges
 - b. No dispersion risk
 - c. Scales linearly with cartridge volume

4.4 Manufacturing & Fabrication Strategy

Phase 4 intentionally avoids high-capital manufacturing.

Fabrication Principles

- 3D-printed structural components
- CNC-cut or moulded housings
- Cartridge-based internal modules
- Off-the-shelf electronics where possible

Assembly Model

- University-assisted prototyping
- NGO-supported regional fabrication
- Localised assembly in deployment regions

4.5 Power & Operations at Scale

Power

- Solar + battery (surface or buoy-linked)
- Duty-cycled subsystems
- Low-power DEP field optimisation

Operations

- No continuous human supervision
- Periodic cartridge swaps
- ROV or surface retrieval as required

4.6 Data, Governance & Trust

Phase 4 embeds:

- University-led monitoring protocols
- NGO-verified environmental reporting
- Transparent performance metrics
- Regulator-friendly documentation

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4.7 Environmental & Social Impact

Phase 4 enables:

- Persistent microplastic interception
- Reduced downstream contamination
- Evidence-based remediation policy
- Local employment in fabrication and maintenance
- Strong ESG alignment for investors

4.8 Phase 4 Success Criteria

Phase 4 is successful if:

- Arrays outperform single-unit deployments
- Capture efficiency remains stable at scale
- Manufacturing cost per unit decreases
- Environmental safety remains uncompromised
- Deployment attracts institutional interest

4.9 Outputs of Phase 4

- Scalable deployment blueprint
- Cost & performance models
- Array-level datasets
- Government-ready briefing packs

- Investment-grade documentation

Phase 5 — Commercialisation, Government Adoption & Institutional Deployment

Status: Strategic rollout phase

Audience: Governments, regulators, multilateral funders, infrastructure investors

Outcome: Adoption pathways, not prototypes

5.1 Purpose of Phase 5

Phase 5 translates validated technology into **deployable environmental infrastructure**.

- Environmental regulation
- Coastal management
- Water authorities
- Mining oversight
- Climate and biodiversity funding mechanisms

5.2 Adoption Pathways

5.2.1 Academic–Government Route

- University-led validation (Edinburgh, Liverpool, Ghana, Abu Dhabi)
- Peer-reviewed datasets

- Pilot-backed evidence packages
- Government acceptance via scientific legitimacy

5.2.2 NGO–Government Route

- Deployment via trusted NGOs (GYEM, iWatch Africa, river trusts)
- Environmental remediation framing
- Community-facing legitimacy
- Pressure-from-below adoption

5.2.3 Infrastructure–Investor Route

- ESG-aligned remediation infrastructure
- Cost-per-kg pollution removal metrics
- Climate, biodiversity, and Blue Economy funding alignment
- Long-term service contracts

5.3 Product Positioning

The system is positioned as:

A modular, ROV-deployable environmental remediation platform for microplastic interception and water quality restoration